



Vienna University of Technology

- Conducted research on cloud computing and middleware
- Taught courses *Advanced Internet Computing* and *Distributed Systems Technologies* on graduate level
- Supervised a large number of master students in their thesis work

Research and Teaching Assistant

Oct. 2007 - Oct. 2011

Vienna University of Technology

- Conducted research on service-based systems engineering
- Taught various courses as TA

Research Software Engineer

Jul. 2005 - Sep. 2007

Siemens PSE, Vienna

- Worked on industrial research project in the area of telecommunications and network management

ACQUIRED GRANTS

I have successfully acquired grants (personal as well as project grants) totalling **around 2.5 million EUR**:

- WASP Academic PhD Student (2022-2026)
Academic Supervisor
Grant value $\approx 3,600,000$ SEK
(≈ 355.000 EUR)
- Debloating AI Systems (2021), AoA SEED Grant
Co-applicant
Total budget 296,000 SEK (my part $\approx 128,000$ SEK)
(≈ 13.000 EUR)
- FFI TrafCloud (2019 – 2021), VINNOVA
Co-applicant
Chalmers budget 5,500,000 SEK (my part $\approx 3,200,000$ SEK)
(≈ 315.000 EUR)
- VR Starting Grant (2019 – 2022), VR
Sole applicant
Grant value 3,639,000 SEK
(≈ 359.000 EUR)
- WASP Academic PhD Student (2017 – 2021), WASP
Academic Supervisor
Grant value $\approx 3,600,000$ SEK
(≈ 355.000 EUR)
- Chalmers AoA Hiring Startup Package (2017 – 2021), Chalmers Foundation
Sole recipient
Grant value 8,000,000 SEK
(≈ 789.000 EUR)

- Individual Project “MINCA” (2016 – 2019), Swiss National Science Foundation
Sole applicant
Grant value 180,000 CHF, at University of Zurich
($\approx$ 163.000 EUR)
- Technology Transfer Project “DevCloud” (2016 – 2019), Hasler Foundation
Co-applicant
Grant value 50,000 CHF, at University of Zurich
($\approx$ 45.000 EUR)

RESEARCH VISITS

Institute of of High-Performance Computing, Singapore
University of Stuttgart, Germany

Feb./Mar. 2013
Apr. 2010

INVITED TALKS AND SEMINARS

(partial list)

- *Bots in Software Engineering*, Invited Talk at WASP4All Webinar, May 2022
- *Unit testing performance using code microbenchmarks. How far are we?*, Invited Talk at the MongoDB Inc. Performance Seminar, November 2020
- *Performance stability of (public) clouds*, Invited Talk at Vehicle Electronics and Connected Services (VECS, industrial conference), April 2019
- *AWS Lambda and #serverless. What’s all the fuzz about?*, Invited Talk at the Chaos Engineering Days Stockholm, December 2018
- *Software performance testing in a public cloud – how bad is it really?*, Invited Talk at the Metrics Day 2018, October 2018
- *Performance Testing of and in the Cloud*, Invited Lecture at the IDEA League Doctoral School on Engineering Complex Systems and IT with Big data and Information Technology, September 2018
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Seminar at IBM Research, Ireland, November 2017
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Invited Talk at the Sixth International Workshop on Load Testing and Benchmarking of Software Systems (LTB 2017), co-located with ICPE’17, April 2017
- *Your User is the Canary - Practices of and Obstacles to Conducting Experiments in Production*, CHOOSE Forum, University of Zurich, September 2016
- *How Predictable is the Performance of IaaS Instances? Some Insights from Benchmarking EC2, GCE, Azure, and Softlayer*, Open Cloud Day, ZHAW Winterthur, June 2016

- *Software Development for the Cloud – Challenges and Opportunities*, Faculty of Informatics, University of Lugano, February 2016
- *Software Development for the Cloud – Challenges and Opportunities*, VU University Amsterdam, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology Ropar, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology New Delhi, November 2015
- *Software Development for the Cloud – Challenges and Opportunities*, Chalmers, Gothenburg, June 2015
- *Engineering Java Applications for the Infrastructure-as-a-Service Cloud*, software evolution & architecture lab, University of Zurich, Zurich, October 2013
- *Building Applications for the Infrastructure-as-a-Service Cloud with CloudScale*, IMDEA Software Institute, Madrid, April 2013
- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, National University of Singapore, School of Computing, March 2013
- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, University of Illinois Advanced Digital Sciences Center, Singapore, February 2013

AWARDS

- Distinguished Reviewer Award (ICSE 2023)
- Best Reviewer Award (SANER 2022)
- Distinguished Reviewer Award (ICSME 2020)
- Distinguished Reviewer Award (MSR 2020)
- Outstanding Reviewer Award (ICPE 2020)
- Outstanding Reviewer Award (ICPE 2019)
- Best Student Paper Award (Middleware 2016)
- Best Short Paper Award (ICPC 2016)

DEPARTMENT SERVICE

- Unit Manager Software Engineering 2 (Chalmers and the University of Gothenburg, 01.04.2022 - ongoing)
- Master Program Administrator Software Engineering (Chalmers and the University of Gothenburg, 01.03.2021 - 01.08.2021)

- Member of the Faculty Senate Drafting Board (IT Faculty, University of Gothenburg, 2021 - 2024)
- Postdoc Representative at the Department of Informatics, University of Zurich (2015 - 2016)
- Member of the Habilitation Committee of Ivona Brandic (TU Vienna, 2013)
- Member of the Habilitation Committee of Karl Michael Göschka (TU Vienna, 2012)

PROFESSIONAL SERVICE

Organized Events

- Co-Organizer of the 1st and 2nd *Vienna Software Seminar (VSS)* on the Relation of Software Architecture and DevOps/Continuous Delivery. Organizers: Uwe Zdun, Philipp Leitner, Florian Rosenberg, Erik Wittern.
- Member of the Steering Committee of the International Workshop on Quality-Aware DevOps (QUDOS). From 2018.
- Dagstuhl GI Seminar on *Software Performance Engineering in the DevOps World* (September 26th – September 30th 2016, Schloss Dagstuhl, Germany). Co-organized with Andre van Hoorn, Pooyan Jamshidi, and Ingo Weber.
- Half-day *CloudWave Training Event* (September 5th 2016, TU Vienna, Austria). Half-day session at ESOC'16. Co-organized with Andreas Metzger and Eliot Salant.

Editorial Memberships

- Associate Editor of Springer Empirical Software Engineering (EMSE)
- Academic Editor of PeerJ Computer Science
- Editorial Board Member of Elsevier's Journal on Systems and Software (JSS). Responsible for coordinating and implementing the publicity efforts of the journal. 2017 - 2019.
- Special Issue Editor for "Converging Fog and Cloud Computing", in IEEE Cloud Computing. Co-edited with Erik Elmroth, Stefan Schulte, and Srikumar Venugopal.
- Special Issue Editor for "Software Tools and Technologies for Monitoring and Prediction of Cloud Services", in Software: Practice and Experience (published by Wiley-Blackwell). Co-edited with Rajiv Ranjan, Rajkumar Buyya, Armin Haller, and Stefan Tai

Chairmanships

- Chair of the Selection Committee of the 2022 SPEC Kaivalya Dixit Distinguished Dissertation Award
- Program Co-Chair of 13th ACM/SPEC International Conference on Performance Engineering (ICPE)
- Awards Co-Chair of 12th ACM/SPEC International Conference on Performance Engineering (ICPE)

- Publication Chair of 10th ACM/SPEC International Conference on Performance Engineering (ICPE)
- PC Co-Chair of the 3rd International Workshop on Quality-Aware DevOps (QUDOS), co-located with the 8th ACM/SPEC International Conference on Performance Engineering (ICPE 2017)
- Co-Chair of a Special Session on Services Computing and Internet Technologies (SerCo 2016) at the International Conference on High Performance Computing & Simulation (HPCS 2016)
- Tutorial Chair of the 16th International Conference on Web Engineering (ICWE2016)
- Workshop Chair of the European Conference on Service-Oriented and Cloud Computing (ESOCC) 2015.
- General Chair of the 1st International Workshop on Monitoring and Prediction of Cloud Services (MoPoC'12), co-located with ICSOC'12.

Reviews for Funding Agencies

(partial list)

- Dutch Research Council (NWO)
- Natural Sciences and Engineering Research Council of Canada (NSERC)
- Deutscher Akademischer Austauschdienst (DAAD)
- Luxembourg National Research Fund (FNR)
- Wallenberg AI, Autonomous Systems, and Software Program (WASP)

Program Committee Memberships

(partial list)

- International Conference on Fundamentals of Software Engineering (FSE 2024)
- International Conference on Software Engineering (ICSE 2023)
- IEEE/ACM International Conference on Automated Software Engineering (ASE 2022 - 2023)
- IEEE International Conference on Cloud Engineering (IC2E 2021)
- European Symposium on Serverless Computing and Applications (ESSCA 2020)
- International Conference on Microservices (MICROSERVICES 2020)
- IEEE/ACM International Conference on Automated Software Engineering - Demonstration Track (ASE 2019)
- IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER, from 2019)
- IEEE International Conference on Software Maintenance and Evolution (ICSME, from 2018)

- International Conference on Mining Software Repositories (MSR, from 2018)
- ACM International Conference on Distributed Event-Based Systems (DEBS 2018)
- ACM/SPEC International Conference on Performance Engineering (ICPE 2018, 2020)
- IEEE International Conference on Cloud Computing (CLOUD, 2015 – 2018)
- International Conference on the Quality of Information and Communications Technology (QUATIC, 2018)
- International Conference on Edge Computing (EDGE 2017)
- International Symposium on Fog and Edge Computing (ISFEC 2017)
- Euromicro Conference on Software Engineering and Advanced Applications (SEAA, 2018)
- International Conference on Service-Oriented Computing (ICSOC, 2015 – 2017)
- European Conference on Service-Oriented and Cloud Computing (ESOCC, 2015, 2017)
- International Conference on Web Engineering (ICWE, 2015, 2019)
- IEEE/ACM International Utility and Cloud Computing Conference (UCC, 2014 – 2017)
- International Conference on Service Oriented Computing and Applications (SOCA, 2014 – ongoing)
- Cloud Challenge, held in conjunction with UCC (2014)
- International Workshop on RESTful Designs (WS-REST, co-located with WWW, 2018)
- International Workshop on Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS, 2018)
- International Workshop on Autonomous Control for Performance and Reliability Trade-offs in Internet of Services (ACPROSS, 2017)
- International Workshop on Inter-Organizational Processes (IOPs, 2016)
- International Workshop on Big Data Software Engineering (BIGDSE, 2015 – ongoing)
- EAI International Conference on Cloud, Networking for IoT Systems (CN4IoT, 2015)
- International Conference on Internet and Web Applications and Services (ICIW, 2014 – 2015)
- International Conference on the Future Internet of Things and Cloud (FiCloud, 2014)
- International Conference on Service Oriented Computing (ICSOC) Demonstration Track (2014 – ongoing)
- International Workshop on Federative and Interoperable Cloud Infrastructures (FedICI, 2014)
- International Workshop on Engineering Cloud Applications & Services (EnCASE, 2014)
- Central-European Workshop on Services and their Composition (ZEUS, 2012 – ongoing)

- International Workshop on Evolutionary Business Processes EVL-BP), co-located with EDOC (2011 – 2013)
- International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE (2013)
- Track on Provisioning and Management of Service Oriented Architecture and Cloud Computing (PROMASC), a Track of the WETICE (2011 – 2012)
- International Workshop on Performance Assessment and Auditing in Service Computing (PAASC), co-located with ICSOC (2011 – 2012)
- Workshop on Emerging Web Services Technology (WEWST), co-located with ECOWS (2009 – 2011)
- International Workshop on Dynamic and Declarative Business Processes (DDBP), co-located with EDOC (2010)

Reviews for Journals

(partial list)

- ACM Transactions on Software Engineering and Methodology (TOSEM)
- Elsevier Information and Software Technology (IST)
- IEEE Transactions on Software Engineering (TSE)
- Journal of Systems and Software (JSS)
- Springer Empirical Software Engineering (ESEM)
- ACM Transactions on Internet Technology (TOIT)
- IEEE Transactions on Cloud Computing (TCC)
- Wiley Concurrency and Computation: Practice and Experience
- Oxford Press: The Computer Journal
- International Journal of Communication Networks and Distributed Systems (IJCND)
- Elsevier Information Processing Letters (IPL)
- IEEE Internet Computing (IC)
- IEEE Software
- Elsevier Journal of Parallel and Distributed Computing (JPDC)
- ACM Transactions on the Web (TWEB)
- Elsevier Journal of Systems and Software (JSS)
- Springer Computing (Archives for Scientific Computing)

- Journal of Computer Science and Technology (JCST)
- International Journal of Cooperative Information Systems (IJCIS)
- IEEE Transactions on Services Computing (TSC)
- Transactions on Pattern Languages of Programming (TPLoP)

RESEARCH STUDENTS

Current PhD Students

- **Ranim Khojah** (Large Language Models in Software Engineering, *co-supervisor*)
- **Huaifeng Zhang** (Debloating ML Systems, *co-supervisor*)
- **Hazem (Peter) Samoaa** (ML for Performance Prediction)
- **Hamdy Michael Ayas** (Architectural Decision Making in Cloud Computing)
- **Linda Erlenhov** (Bots in Software Development)

Graduated PhD Students

- **Joel Scheuner** (Cloud Benchmarking, now software engineer at LocalStack, Switzerland)
- **Christoph Laaber** (UZH, microbenchmarking in CI builds, co-supervised with Harald C. Gall, now postdoc at Simula, Norway))
- **Gerald Schermann** (UHZ, continuous experimentation, co-supervised with Harald C. Gall, now release engineer at digitec.ch, Switzerland)
- **Jürgen Cito** (UZH, cloud performance engineering, co-supervised with Harald C. Gall, now Assistant Professor at TU Wien, Austria)

PhD Committee Appointments

- **César Soto-Valero** (KTH, member of grading committee)
- **Mohammad Reza Saleh Sedghpour** (Umeå University, member of grading committee)
- **Alim Ul Gias** (Imperial College London, external examiner)
- **Luca Traini** (Università degli Studi dell'Aquila, assessor)
- **Hugo Sica de Andrade** (Chalmers University of Technology, substitute member of grading committee)
- **Lars Larsson** (Umeå University, faculty opponent)
- **Alfred Åkesson** (Lund University, licenciate discussion leader)

FORMAL PEDAGOGICAL TRAINING

- Diversity and inclusion in higher education (2018, 2 ECTS)
- University teaching and learning (2017, 3 ECTS)
- Enhancing learning through writing (2018, 5 ECTS)
- Perspectives on learning (2018, 2.5 ECTS)
- Supervision of research (2017, 3 ECTS)

TEACHING ACTIVITIES

Teaching at Chalmers | GU

- Distributed Systems Mini-Project (DIT356) (since 2023, $\approx$ 80 students)
- Architectures of Scale-Out Systems (DAT490) (since 2022, $\approx$ 40 students)
- Mobile and Web Development (DIT341) (2019 – 2021, $\approx$ 100 students)
- Data Management (DIT032/DAT335) (2018 – 2021, $\approx$ 150 students)

Previous Teaching at University of Zurich

- Informatik 1 (fall 2014 – 2016, $\approx$ 250 students)
- Advanced Software Engineering (spring 2014 – 2017, 10 – 20 students)
- Seminar Advanced Software Engineering (fall 2015 – 2016, 10 – 20 students)
- Master Basismodul (2014 – 2017)
- Informatik-Vertiefung (2014 – 2017)

Previous Teaching at TU Vienna

- Advanced Internet Computing (fall 2008 - 2013, 140 students)
- Distributed Systems Technologies (spring 2009 - 2013, 140 students)
- Internet Computing Lab Work (spring 2009 - 2013)
- Verteilte Systeme (fall 2009 - fall 2010, $\approx$ 400 students)
- Grid Computing (spring 2008 - spring 2009, 10 students)

Thesis Supervision at Chalmers | University of Gothenburg

(partial list, total ≈ 20 supervised theses)

- Albin Johansson and Carl Holmberg: The Impact of Compiler Warnings on Code Quality in C++ Projects (master's thesis, 2023)
- Florian Johan Markusse: The impact of benchmarking bots on Open-Source-Software projects (master's thesis, co-supervisor, TU Eindhoven, 2021)
- Niclas Alexandersson: JUnit-to-JMH - Automated Generation of Performance Benchmarks from Existing Unit Tests in Java (master's thesis, 2021)
- Erik Bäcknas and Eric Börjesson: Total Cost of Ownership Framework to Estimate the Cost of Cloud Computing Services (master's thesis, 2021)
- Yan Wang and Mihkel Sildnik: Debloating Machine Learning Systems (master's thesis, co-advisor, 2021)
- Joel Sanderöd Roxell and Matthias Lundell: Automatic container profiling and resource prediction using container orchestration (master's thesis, 2019)
- Yunfang Guo: The Impact of Adopting Continuous Integration on the Delivery Time of Pull Requests - A Partial Replication and Extension (master's thesis, 2019)
- Viktor Holmqvist and Jonathan Nilsfors: Cachematic - Automatic Invalidation in Application-Level Caching Systems (master's thesis, 2018)

ACADEMIC JOURNAL PAPERS [scholar]

- H. M. Ayas, **P. Leitner**, and R. Hebig: *An Empirical Study of the Systemic and Technical Migration Towards Microservices*, Empirical Software Engineering (EMSE), 2023. To appear.
 - M. Grambow, D. Kovalev, C. Laaber, **P. Leitner**, and D. Bermbach: *Using Microbenchmark Suites to Detect Application Performance Changes*, IEEE Transactions on Cloud Computing (TCC), 2022. To appear.
- M. Jangali, Y. Tang, N. Alexandersson, **P. Leitner**, J. Yang, and W. Shang: *Automated Generation and Evaluation of JMH Microbenchmark Suites from Unit Tests*, IEEE Transactions on Software Engineering (TSE), 2022. To appear.
- F. Markusse, **P. Leitner**, and A. Serebrenik: *Using Benchmarking Bots for Continuous Performance Assessment*, IEEE Software, 2022. To appear.
 - H. Samoaa, F. Bayram, P. Salza, and **P. Leitner**: *A Systematic Mapping Study of Source Code Representation for Deep Learning in Software Engineering*, IET Software, 2022.
 - L. Erlenhov, F. Gomes, and **P. Leitner**: *Dependency management bots in open-source systems—prevalence and adoption*. PeerJ Computer Science 8:e849 (2022). <https://doi.org/10.7717/peerj-cs.849>
- C. Laaber, **P. Leitner**, and H. Gall: *Applying Test Case Prioritization to Software Microbenchmarks*, Empirical Software Engineering, 26, 133 (2021). <https://doi.org/10.1007/s10664-021-10037-x>
- M. Grambow, C. Laaber, **P. Leitner**, and D. Bermbach: *Using Application Benchmark Call Graphs to Quantify and Improve the Practical Relevance of Microbenchmark Suites*, PeerJ Computer Science 7:e548 (2021). <https://doi.org/10.7717/peerj-cs.548>
 - J. Scheuner and **P. Leitner**: *Function-as-a-Service Performance Evaluation: A Multivocal Literature Review*, Journal of Systems and Software (JSS), vol. 170. 2020.
 - Y. Guo and **P. Leitner**: *Studying the Impact of CI on Pull Request Delivery Time in Open Source Projects — a Conceptual Replication*, PeerJ Computer Science, 2019.
- D. Costa, CP. Bezemer, **P. Leitner**, and A. Andrzejak: *What’s Wrong With My Benchmark Results? Studying Bad Practices in JMH Benchmarks*, IEEE Transactions on Software Engineering (TSE). Early access, doi: 10.1109/TSE.2019.2925345.
- B. Varghese, **P. Leitner**, S. Ray, K. Chard, A. Barker, Y. Elkhatib, H. Herry, C.H. Hong, J. Singer, F.P. Tso, E. Yoneki, and M.F. Zhani: *Cloud Futurology*, IEEE Computer 52(9): 68-77 (2019)

- C. Laaber, J. Scheuner, and **P. Leitner**: *Software Microbenchmarking in the Cloud. How Bad is it Really?*, Empirical Software Engineering 24(4): 2469-2508 (2019)

- **P. Leitner**, E. Wittern, J. Spillner, and W. Hummer: *A Mixed-Method Empirical Study of Function-as-a-Service Software Development in Industrial Practice*, Journal of Systems and Software 149: 340-359 (2019)
- G. Schermann, J. Cito, **P. Leitner**, U. Zdun, and H. Gall: *We're Doing It Live: A Multi-Method Empirical Study on Continuous Experimentation*, Information and Software Technology 99: 41-57 (2018)
- G. Schermann, J. Cito, and **P. Leitner**: *Continuous Experimentation - Challenges, Implementation Techniques, and Current Research*, IEEE Software 35(2): 26-31 (2018)
- O. Skarlat, M. Nardelli, S. Schulte, M. Borkowski, and **P. Leitner**: *Optimized IoT Service Placement in the Fog*, Service Oriented Computing and Applications 11(4): 427-443 (2017)

- **P. Leitner** and J. Cito: *Patterns in the Chaos – a Study of Performance Variation and Predictability in Public IaaS Clouds*, ACM Transactions on Internet Technology (TOIT), vol. 16, iss. 3, 2016.

- R. Zabolotnyi, **P. Leitner**, W. Hummer, and S. Dustdar: *JCloudScale: Closing the Gap Between IaaS and PaaS*, ACM Transactions on Internet Technology (TOIT), vol. 15, iss. 3, p. 10:1–10:20, 2015.
- J. Cito, D. Gotowka, **P. Leitner**, R. Pelette, D. Suljoti, and S. Dustdar: *Identifying Web Performance Degradations through Synthetic and Real-User Monitoring*, Journal of Web Engineering, 2015.
- A. Metzger, **P. Leitner**, D. Ivanovic, E. Schmieders, R. Franklin, M. Carro, S. Dustdar, and K. Pohl: *Comparing and Combining Predictive Business Process Monitoring Techniques*, IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 45, no. 2, pp. 276 - 290. 2014.
- C. Inzinger, W. Hummer, B. Satzger, **P. Leitner**, and S. Dustdar: *Generic Event-Based Monitoring and Adaptation Methodology for Heterogeneous Distributed Systems*, Software: Practice and Experience, 2014.
- **P. Leitner**, W. Hummer and S. Dustdar: *Data-Driven and Automated Prediction of Service Level Agreement Violations in Service Compositions*, Distributed and Parallel Databases (Springer), vol. 31, no. 3, pp. 447 - 470. 2013
- W. Hummer, O. Raz, O. Shehory, **P. Leitner**, and S. Dustdar: *Testing of Data-Centric and Event-Based Dynamic Service Compositions*, Software Testing, Verification and Reliability, vol. 23, iss. 6, pp. 465-497, 2013.
- **P. Leitner**, W. Hummer and S. Dustdar: *Cost-Based Optimization of Service Compositions*, IEEE Transactions on Services Computing (TSC). 2011, doi:10.1109/TSC.2011.53

- W. Hummer, **P. Leitner** and S. Dustdar: *SEPL - A Domain-Specific Language and Execution Environment for Protocols of Stateful Web Services*, Distributed and Parallel Databases (DPD), vol. 29, nr. 4, pp. 277-307. 2011, doi:10.1007/s10619-011-7079-6
- B. Wetzstein, **P. Leitner**, F. Rosenberg, S. Dustdar and F. Leymann: *Identifying Influential Factors of Business Process Performance Using Dependency Analysis*, Enterprise Information Systems (EIS), vol. 5, nr. 1. 2011, doi:10.1080/17517575.2010.493956
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *End-to-End Support for QoS-Aware Service Selection, Binding and Mediation in VRESCo*, IEEE Transactions on Services Computing (TSC), vol. 3, nr. 3, pp. 193-205. 2010, doi:10.1109/TSC.2010.20
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *Selective Service Provenance in the VRESCo Runtime*, International Journal on Web Services Research (JWSR), vol 7, no. 2, pp. 65-86, 2010, doi: 10.4018/jwsr.2010040104
- **P. Leitner**, F. Rosenberg, S. Dustdar, Daios: *Efficient Dynamic Web Service Invocation*, IEEE Internet Computing, vol. 13, no. 3, pp. 72-80, May/June 2009, doi:10.1109/MIC.2009.57

ACADEMIC CONFERENCE PAPERS [\[scholar\]](#)

- J. Scheuner, R. Deng, J.-P. Steghöfer and **P. Leitner**: *CrossFit: Fine-grained Benchmarking of Serverless Application Performance across Cloud Providers*, in Proceedings of the 15th IEEE/ACM International Conference on Utility and Cloud Computing (UCC). 2022.
- H. Samoa, A. Longa, M. Mohamad, M. H. Chehrehghani, and **P. Leitner**: *TEP-GNN: Accurate Execution Time Prediction of Functional Tests using Graph Neural Networks*, in Proceedings of the 22nd International Conference on Product-Focused Software Process Improvement (PROFES). 2022.
- J. Scheuner, M. Bertilsson, O. Gronqvist, H. Tao, H. Lagergren, JP. Steghöfer and **P. Leitner**: *TriggerBench: A Performance Benchmark for Serverless Function Triggers* (Short Paper), in Proceedings of the International Conference on Cloud Engineering (IC2EE). 2022.
- H. M. Ayas, F. Gomes, and **P. Leitner**: *Empirical Analysis of Microservices Systems Using Consumer-Driven Contract Testing*, in Proceedings of the Euromicro Conference on Software Engineering and Advanced Applications (SEAA). 2022.
- H. M. Ayas, **P. Leitner**, and R. Hebig: *A Grounded Theory study on the migration journey towards microservices*, in Proceedings of the 22nd International Conference on Product-Focused Software Process Improvement (PROFES). 2021.
- H. M. Ayas, R. Hebig, and **P. Leitner**: *Facing the Giant: a Grounded Theory Study of Decision-Making in Microservices Migrations*, in Proceedings of the 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM). 2021.
- P. Samoa and **P. Leitner**: *An Exploratory Study of the Impact of Parameterization on JMH Measurement Results in Open-Source Projects*, in Proceedings of the 2021 ACM/SPEC International Conference on Performance Engineering (ICPE). 2021.

- C. Laaber, S. Würsten, H. C. Gall, and **P. Leitner**: *Dynamically Reconfiguring Software Microbenchmarks: Reducing Execution Time Without Sacrificing Result Quality*, in Proceedings of the 2020 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE). 2020.
- L. Erlenhov, F. Gomes, and **P. Leitner**: *An Empirical Study of Bots in Software Development - Characteristics and Challenges from a Practitioner's Perspective*, in Proceedings of the 2020 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE). 2020.
- G. Schermann, F. Oliveira, E. Wittern, and **P. Leitner**: *Topology-aware Continuous Experimentation in Microservice-based Applications*, in Proceedings of the 2020 International Conference on Service Oriented Computing (ICSOC). 2020.
- J. Cito, **P. Leitner**, M. Rinard, and H. C. Gall: *Interactive Production Performance Feedback in the IDE*, in Proceedings of the 41st ACM/IEEE International Conference on Software Engineering (ICSE). 2019.
- V. Holmqvist, J. Nilsfors, and **P. Leitner**: *Cachematic – Automatic Invalidiation in Application-Level Caching Systems*, in Proceedings of the 2019 ACM/SPEC International Conference on Performance Engineering (ICPE) – Industry Track. 2019.
- G. Schermann and **P. Leitner**: *Search-Based Scheduling of Experiments in Continuous Deployment*, in Proceedings of the 34rd IEEE International Conference on Software Maintenance and Evolution (ICSME). 2018.
- C. Laaber and **P. Leitner**: *An Evaluation of Open-Source Software Microbenchmark Suites for Continuous Performance Assessment*, in Proceedings of the 15th International Conference on Mining Software Repositories (MSR), 2018.
- C. Vassallo, G. Schermann, F. Zampetti, D. Romano, **P. Leitner**, A. Zaidman, M. Di Penta, and S. Panichella: *A Tale of CI Build Failures: an Open Source and a Financial Organization Perspective*, in Proceedings of the 33rd IEEE International Conference on Software Maintenance and Evolution (ICSME). 2017
- G. Mazlamo, J. Cito, and **P. Leitner**: *Extraction of Microservices from Monolithic Software Architectures*, in Proceedings of the 24th IEEE International Conference on Web Services (ICWS) - Applications Track. 2017
- J. Cito, G. Schermann, E. Wittern, **P. Leitner**, S. Zumberi, and H. C. Gall: *An Empirical Analysis of the Docker Container Ecosystem on GitHub*, in Proceedings of the 14th International Conference on Mining Software Repositories (MSR), 2017.
- T. Rausch, W. Hummer, **P. Leitner**, and S. Schulte: *An Empirical Analysis of Build Failures in the Continuous Integration Workflows of Java-Based Open-Source Software*, in Proceedings of the 14th International Conference on Mining Software Repositories (MSR), 2017.

- C. Davatz, C. Inzinger, J. Scheuner, and **P. Leitner** : *An Approach and Case Study of Cloud Instance Type Selection for Multi-Tier Web Applications*, in Proceedings of the 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), 2017.
- J. Cito, F. Oliveira, **P. Leitner**, P. Nagpurkar, and H. C. Gall: *Context-Based Analytics – Establishing Explicit Links between Runtime Traces and Source Code*, In Proceedings of the 38th International Conference on Software Engineering (ICSE'17) – Software Engineering in Practice (SEIP) Track, 2017.
- **P. Leitner** and CP. Bezemer: *An Exploratory Study of the State of Practice of Performance Testing in Java-Based Open Source Projects*, in Proceedings of the 7th ACM/SPEC International Conference on Performance Engineering (ICPE), 2017.

- G. Schermann, D. Schöni, **P. Leitner**, and H. C. Gall: *Bifrost – Supporting Continuous Deployment with Automated Enactment of Multi-Phase Live Testing Strategies*, in Proceedings of the 2016 ACM/IFIP/USENIX Middleware Conference. **Best Student Paper Award**.

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